

Supplementary Information

Adaptive Weighted and Hybrid Extensions of the Naive Bayes Enrichment Method for Mixed-Type Data: An Evaluation within the NB/NBEM Family

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This file contains the supplementary tables and algorithmic descriptions referenced in the manuscript. Numerical results correspond to the reported analysis; the five-seed in-family tables use the primary Bank Marketing protocol with `duration` excluded, and the focal external comparison uses the same locked twenty-seed plan for all four focal models.

Supplementary Table S1. Implementation-level complexity summary for the evaluated NBEM extensions.

Method	Training complexity	Main additional operation
NBEM-prop	$O(nmK)$	Type-specific Naive Bayes estimation
Adaptive Weighted NBEM	$O(nmK + nRK)$	Entropy-margin scores and Softmax gating
Hybrid NBEM	$O(nmK + nq^2 + ns + EnP + InRK^2)$	Capped interactions, MLP learning, and logistic stacking

Supplementary Table S2. Fixed model configurations used in the reported experiments.

Model or stage	Configuration
Bernoulli / Categorical NB	Laplace smoothing $\alpha = 1$; posterior columns aligned to the global class order.
Gaussian NB	Default scikit-learn variance smoothing; numerical inputs standardized using training-fold statistics.
WNB	Categorical NB on the discretized representation with mutual-information feature weights estimated only from the training fold.
NBEM-prop	Bernoulli, categorical, and Gaussian NB components; each available component weighted in proportion to the number of predictors in its feature group.
Adaptive Weighted NBEM	Entropy-margin score with $\lambda = 0.5$ and Softmax temperature $\tau = 1$.
Dependency-aware component	Five-bin training-fold discretization; at most 80 mutual-information candidates; 10 pairwise interactions selected by absolute correlation.
Probabilistic deep component	MLP hidden layers (128, 64); ReLU; Adam; L_2 penalty 10^{-4} ; initial learning rate 10^{-3} ; adaptive schedule; max 300 iterations; early stopping with validation fraction 0.1 and patience 20.
Hybrid meta-learner	Logistic regression over concatenated component posteriors; max 1000 iterations.
Logistic Regression baseline	Fold-local one-hot categorical/Boolean representation; standardized numerical features; $C = 1$; <code>lbfgs</code> ; max 2000 iterations.
Random Forest baseline	100 trees; Gini criterion; unrestricted depth; minimum split size 2; minimum leaf size 1; <code>max_features=sqrt</code> ; bootstrap enabled.
Randomization	Five original in-family seeds {13, 21, 42, 87, 123}; locked twenty-seed focal comparison for Hybrid/NBEM-prop/Logistic/RF; reference seed 42 for ablation.

Supplementary Table S3. Dataset-wise weighted F1-score averaged over the five predefined seeds under the primary Bank Marketing protocol with duration excluded.

Dataset	WNB	NBEM-prop	Adaptive Weighted NBEM	Hybrid NBEM
Abalone	0.1971	0.2095	0.2089	0.2485
Bank	0.8282	0.8491	0.8449	0.8668
Car Evaluation	0.5768	0.8380	0.8380	0.9398
Fetal Health	0.6815	0.8316	0.8460	0.9240
Kidney Disease	0.9817	0.9265	0.9486	0.9825
Cirrhosis Survival	0.6736	0.6829	0.6804	0.7307
Credit Approval	0.8609	0.7665	0.7749	0.8600
Diabetes	0.3788	0.4791	0.4768	0.5355
Dry Beans	0.8032	0.8971	0.8971	0.9299
E. coli	0.6445	0.8293	0.8307	0.8250
Glass Identification	0.5692	0.4245	0.4245	0.6500
Haberman Survival	0.6231	0.6926	0.6926	0.6910
Heart Disease	0.3801	0.5397	0.4969	0.5441
Heart Failure	0.8011	0.7227	0.7156	0.8266
Hepatitis	0.7027	0.8384	0.8402	0.8323
Indian Liver Patients	0.5943	0.5594	0.5647	0.6350
Internet Ads	0.7953	0.9605	0.9648	0.9679
Mushroom	0.9852	0.9429	0.9541	0.9997
Students	0.6359	0.6525	0.6648	0.7505
Productivity Prediction	0.4949	0.5194	0.5054	0.7033

Supplementary Table S4. Dataset-wise Macro-F1 averaged over the five predefined seeds under the primary Bank Marketing protocol with duration excluded.

Dataset	WNB	NBEM-prop	Adaptive Weighted NBEM	Hybrid NBEM
Abalone	0.0760	0.1208	0.1192	0.1404
Bank	0.4689	0.5575	0.5401	0.6231
Car Evaluation	0.2059	0.6053	0.6053	0.8618
Fetal Health	0.2918	0.7004	0.7158	0.8583
Kidney Disease	0.9803	0.9235	0.9462	0.9813
Cirrhosis Survival	0.4652	0.4924	0.4858	0.5173
Credit Approval	0.8591	0.7592	0.7681	0.8581
Diabetes	0.2345	0.3382	0.3520	0.3966
Dry Beans	0.7034	0.9064	0.9064	0.9410
E. coli	0.2810	0.5247	0.5258	0.5034
Glass Identification	0.3560	0.4579	0.4579	0.5457
Haberman Survival	0.4237	0.5566	0.5566	0.5583
Heart Disease	0.1404	0.2929	0.2521	0.2928
Heart Failure	0.7559	0.6590	0.6490	0.7953
Hepatitis	0.4424	0.7585	0.7564	0.7256
Indian Liver Patients	0.4164	0.5579	0.5620	0.5014
Internet Ads	0.4624	0.9145	0.9239	0.9328
Mushroom	0.9852	0.9427	0.9540	0.9997
Students	0.5114	0.5803	0.5870	0.6843
Productivity Prediction	0.3002	0.3672	0.3403	0.5591

Supplementary Table S5. Dataset-wise weighted F1-score for the focal four-model comparison, averaged over the same twenty locked seeds.

Dataset	NBEM-prop	Hybrid NBEM	Logistic Regression	Random Forest
Abalone	0.2094	0.2503	0.2448	0.2346
Bank	0.8491	0.8669	0.8645	0.8716
Car Evaluation	0.8389	0.9413	0.9066	0.9644
Fetal Health	0.8310	0.9214	0.8918	0.9452
Kidney Disease	0.9275	0.9831	0.9896	0.9933
Cirrhosis Survival	0.6838	0.7352	0.7355	0.7367
Credit Approval	0.7670	0.8602	0.8598	0.8740
Diabetes	0.4792	0.5362	0.5249	0.5246
Dry Beans	0.8971	0.9300	0.9242	0.9246
E. coli	0.8383	0.8222	0.8644	0.8503
Glass Identification	0.4234	0.6510	0.6070	0.7699
Haberman Survival	0.6934	0.6924	0.6764	0.6720
Heart Disease	0.5373	0.5454	0.5645	0.5166
Heart Failure	0.7203	0.8249	0.8305	0.8434
Hepatitis	0.8374	0.8295	0.8466	0.8269
Indian Liver Patients	0.5585	0.6314	0.6746	0.6881
Internet Ads	0.9606	0.9692	0.9775	0.9784
Mushroom	0.9429	0.9997	0.9999	1.0000
Students	0.6533	0.7522	0.7484	0.7599
Productivity Prediction	0.5174	0.7046	0.6187	0.7721

Supplementary Table S6. Dataset-wise Macro-F1 for the focal four-model comparison, averaged over the same twenty locked seeds.

Dataset	NBEM-prop	Hybrid NBEM	Logistic Regression	Random Forest
Abalone	0.1210	0.1401	0.1378	0.1419
Bank	0.5576	0.6237	0.6113	0.6417
Car Evaluation	0.6065	0.8660	0.7687	0.8990
Fetal Health	0.7001	0.8514	0.7925	0.8997
Kidney Disease	0.9245	0.9820	0.9889	0.9929
Cirrhosis Survival	0.4929	0.5213	0.5401	0.5162
Credit Approval	0.7599	0.8584	0.8585	0.8726
Diabetes	0.3383	0.3976	0.3840	0.3724
Dry Beans	0.9065	0.9412	0.9359	0.9357
E. coli	0.5428	0.5118	0.6310	0.6165
Glass Identification	0.4574	0.5461	0.5296	0.7176
Haberman Survival	0.5581	0.5594	0.5269	0.5572
Heart Disease	0.2892	0.2932	0.3187	0.2733
Heart Failure	0.6561	0.7935	0.8021	0.8168
Hepatitis	0.7548	0.7196	0.7549	0.7229
Indian Liver Patients	0.5574	0.4963	0.5629	0.5960
Internet Ads	0.9148	0.9353	0.9521	0.9542
Mushroom	0.9427	0.9997	0.9999	1.0000
Students	0.5816	0.6864	0.6762	0.6948
Productivity Prediction	0.3641	0.5593	0.4709	0.6518

Supplementary Table S7. Dataset-wise weighted Precision (Recall), averaged over the five pre-defined seeds under the primary Bank Marketing protocol with duration excluded.

Dataset	WNB	NBEM-prop	Adaptive Weighted NBEM	Hybrid NBEM
Abalone	0.170(0.246)	0.199(0.242)	0.195(0.241)	0.239(0.275)
Bank	0.780(0.883)	0.852(0.885)	0.849(0.884)	0.869(0.892)
Car Evaluation	0.490(0.700)	0.848(0.850)	0.848(0.850)	0.943(0.941)
Fetal Health	0.606(0.778)	0.873(0.815)	0.876(0.833)	0.925(0.925)
Kidney Disease	0.983(0.982)	0.940(0.926)	0.956(0.948)	0.983(0.982)
Cirrhosis Survival	0.709(0.717)	0.696(0.704)	0.693(0.708)	0.719(0.757)
Credit Approval	0.865(0.861)	0.804(0.778)	0.808(0.785)	0.864(0.861)
Diabetes	0.483(0.539)	0.533(0.565)	0.524(0.562)	0.549(0.581)
Dry Beans	0.795(0.819)	0.899(0.897)	0.899(0.897)	0.930(0.930)
E. coli	0.593(0.722)	0.826(0.838)	0.827(0.840)	0.818(0.840)
Glass Identification	0.544(0.630)	0.474(0.462)	0.474(0.462)	0.634(0.690)
Haberman Survival	0.541(0.735)	0.705(0.746)	0.705(0.746)	0.710(0.740)
Heart Disease	0.293(0.541)	0.526(0.566)	0.466(0.560)	0.522(0.587)
Heart Failure	0.832(0.821)	0.750(0.751)	0.745(0.747)	0.835(0.832)
Hepatitis	0.631(0.794)	0.856(0.837)	0.854(0.842)	0.842(0.846)
Indian Liver Patients	0.509(0.714)	0.795(0.559)	0.795(0.564)	0.623(0.699)
Internet Ads	0.740(0.860)	0.962(0.962)	0.966(0.966)	0.969(0.968)
Mushroom	0.986(0.985)	0.948(0.943)	0.957(0.954)	1.000(1.000)
Students	0.586(0.709)	0.655(0.668)	0.665(0.688)	0.748(0.759)
Productivity Prediction	0.524(0.613)	0.575(0.621)	0.563(0.614)	0.707(0.729)

Supplementary Table S8. Model-only timing in the twenty-seed focal runs. Values are equal-dataset means in the measured software/hardware environment; preprocessing time is excluded.

Model	Model fit (s)	Inference (s)
NBEM-prop	0.044	0.018
Hybrid NBEM	34.005	0.110
Logistic Regression	5.066	0.003
Random Forest	5.183	0.305

Supplementary Table S9. Five-seed in-family Wilcoxon comparisons. Holm correction is applied within each F1 metric across the three planned comparators.

Metric	Comparator	Mean comparator	Mean Hybrid	Holm-adjusted p
Weighted F1	WNB	0.6604	0.7722	1.717e-05
Weighted F1	NBEM-prop	0.7081	0.7722	7.248e-05
Weighted F1	Adaptive Weighted NBEM	0.7085	0.7722	7.248e-05
Macro-F1	WNB	0.4680	0.6638	1.144e-05
Macro-F1	NBEM-prop	0.6008	0.6638	1.701e-03
Macro-F1	Adaptive Weighted NBEM	0.6002	0.6638	1.701e-03

Supplementary Table S10. Twenty-seed focal paired comparisons using one twenty-seed mean per dataset as the inferential unit. Holm correction is across the three planned comparator contrasts within each metric.

Metric	Comparator	Mean difference	95% bootstrap CI	W/T/L	Rank-biserial	Holm p
Weighted F1	NBEM-prop	+0.0641	[0.0391, 0.0920]	17/0/3	+0.924	0.000143
Weighted F1	Logistic Regression	+0.0048	[-0.0069, 0.0177]	11/0/9	+0.143	0.596
Weighted F1	Random Forest	-0.0150	[-0.0308, -0.0019]	6/0/14	-0.448	0.165
Macro-F1	NBEM-prop	+0.0628	[0.0304, 0.0978]	17/0/3	+0.752	0.00596
Macro-F1	Logistic Regression	+0.0020	[-0.0188, 0.0222]	10/0/10	+0.029	0.927
Macro-F1	Random Forest	-0.0295	[-0.0526, -0.0100]	5/0/15	-0.619	0.0272

Supplementary Table S11. Actual cross-fitted Hybrid NBEM robustness check. Values are cross-fitted minus main non-cross-fitted means across the original five seeds.

Dataset	Δ Weighted F1	Δ Macro-F1	Δ ROC-AUC
bank marketing	-0.0018	-0.0080	-0.0000
car evaluation	-0.0114	-0.0195	-0.0011
dry bean dataset	-0.0010	-0.0009	-0.0001
heart disease	+0.0090	+0.0285	+0.0069
internet advertisements	+0.0051	+0.0096	+0.0006
productivity prediction of garment employees	-0.0047	-0.0263	+0.0002

Supplementary Table S12. Cross-validation sensitivity analysis for Hybrid NBEM. Values are mean $\pm$ standard deviation across the original five seeds.

Dataset	CV protocol	Weighted F1	Macro-F1
Abalone	KFold_10	0.2485 $\pm$ 0.0055	0.1404 $\pm$ 0.0033
Abalone	KFold_5	0.2465 $\pm$ 0.0058	0.1276 $\pm$ 0.0045
Fetal Health	StratifiedKFold_10	0.9240 $\pm$ 0.0064	0.8583 $\pm$ 0.0141
Fetal Health	StratifiedKFold_5	0.9191 $\pm$ 0.0068	0.8478 $\pm$ 0.0119
Diabetes	StratifiedGroupKFold_10	0.5355 $\pm$ 0.0020	0.3966 $\pm$ 0.0024
Diabetes	StratifiedKFold_10	0.5359 $\pm$ 0.0010	0.3976 $\pm$ 0.0011
E. coli	RepeatedStratifiedKFold_2x3	0.8161 $\pm$ 0.0136	0.4937 $\pm$ 0.0211
E. coli	StratifiedKFold_2	0.8250 $\pm$ 0.0256	0.5034 $\pm$ 0.0425
Glass Identification	RepeatedStratifiedKFold_9x3	0.6409 $\pm$ 0.0170	0.5368 $\pm$ 0.0210
Glass Identification	StratifiedKFold_5	0.6569 $\pm$ 0.0104	0.5636 $\pm$ 0.0218
Productivity Prediction	StratifiedKFold_10	0.7033 $\pm$ 0.0101	0.5591 $\pm$ 0.0156
Productivity Prediction	StratifiedKFold_5	0.6928 $\pm$ 0.0129	0.5427 $\pm$ 0.0277

Supplementary Table S13. Duplicate-grouped sensitivity for the two datasets exceeding the predeclared duplicate-exposure threshold. Values are grouped minus record-level means for Hybrid NBEM.

Dataset	Δ Weighted F1	Δ Macro-F1	Δ ROC-AUC
Haberman Survival	-0.0040	-0.0048	-0.0026
Internet Ads	-0.0081	-0.0156	-0.0111

Supplementary Note 1: NBEM pseudo-code

The pseudo-code below reproduces the implementation description retained from the reference-model section. AutoTFS is the automated threshold-learning feature-selection procedure cited

in the main manuscript.

```

NBEM(D, v, weights)

- F = {F1, ..., Fm} <- set of features in D
- W = {w_f1, ..., w_fm} <- set of weights by AutoTFS
- A = {D_b, D_c, D_g} <- Boolean, categorical, and numerical datasets, respectively

- If weights are used:
    W_b, W_c, W_g <- sum_{f_i in D_b} w_fi,
                    sum_{f_i in D_c} w_fi,
                    sum_{f_i in D_g} w_fi
- Else:
    set each W_b, W_c, W_g by uniform distribution

- For each D_i in A:
    add the target variable to D_i
    train NB classifier by the data type

- y_b, y_c, y_g <- prediction of v by the Boolean, categorical, and numerical classifiers
- Return y_hat(D, v)

```

Supplementary Note 2: Adaptive Weighted NBEM pseudo-code

The pseudo-code below summarizes the entropy-margin adaptive weighting mechanism defined in the main manuscript.

```

AdaptiveWeightedNBEM(D, v, lambda, tau)

- Partition the predictors into Boolean, categorical, and numerical groups
- Fit Bernoulli NB, Categorical NB, and Gaussian NB when their groups are nonempty
- For every available component t:
    p_t(v) <- aligned class-posterior vector
    C_t(v) <- 1 - H(p_t(v))/log(K)
    M_t(v) <- largest(p_t(v)) - second_largest(p_t(v))
    g_t(v) <- lambda*C_t(v) + (1-lambda)*M_t(v)
- alpha(v) <- softmax(g(v)/tau)
- p_AW(v) <- sum_t alpha_t(v)*p_t(v)
- Return argmax_k p_AW,k(v)

```